

Gas Compressor Vibration

Field Manual — Setup and Troubleshooting Reference

Use case definition. A gas compressor raises gas pressure for export, reinjection, or gathering. Abnormal behavior includes increased vibration, bearing temperature, unstable suction/discharge pressure, liquid carryover, seal-gas deviation, or anti-surge cycling. Causes include imbalance, misalignment, looseness, lubrication or seal-gas problems, piping forces, liquid carryover, and operating-point instability. Troubleshooting prevents rotor, bearing, seal, and casing damage.

Manual setup and use. Use compressor maps, OEM alarm/trip limits, approved anti-surge logic, and gas-quality requirements. Values are training/reference values.

Safety note. Treat gas release, rotating machinery, and seal-gas systems as hazardous. Do not enter the rotating equipment area or loosen pressurized connections.

Quick-use workflow

Use this sequence for every event: establish a safe operating state; capture the initial numerical readings; compare each value with the stated reference; apply the trigger-based action; correct the identified component or process path; restore operation gradually; and confirm the expected numerical result. If an OEM/site alarm or trip limit differs from a training value, the OEM/site limit governs.

Scenario 1: Bearing overheating

Field Condition

bearing 96 °C from 68; vibration 7.0 mm/s RMS from 2.4; lube oil 1.6 barg from 2.4

Problem

Heat and vibration indicate lubrication, alignment, or bearing distress.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses temperature above 90 °C.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when temperature 65–75 °C and vibration 2–4 mm/s RMS.

Expected Result

temperature 65–75 °C and vibration 2–4 mm/s RMS

Scenario 2: Coupling misalignment

Field Condition

angular offset 0.31 mm/100 mm; parallel 0.38 mm; vibration 6.6 mm/s RMS

Problem

Alignment error is producing 1X vibration and bearing heat.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses offset above 0.10 mm.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.

4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when offset below 0.10 mm and vibration 2–4 mm/s RMS.

Expected Result

offset below 0.10 mm and vibration 2–4 mm/s RMS

Scenario 3: Rotor imbalance

Field Condition

1X vibration 8.2 mm/s RMS; phase shift 14°; speed 7,200 RPM; load 68 %

Problem

Dominant 1X vibration suggests rotor imbalance or deposits.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses 1X above 6 mm/s RMS.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when 1X vibration below 3 mm/s RMS after approved balance/cleaning.

Expected Result

1X vibration below 3 mm/s RMS after approved balance/cleaning

Scenario 4: Lubrication problem

Field Condition

lube pressure 1.1 barg (normal 2.2–2.8); oil 82 °C; bearing 89 °C

Problem

Low oil pressure and high oil temperature reduce bearing film.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses pressure below 1.8 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when pressure 2.2–2.8 barg and oil 45–60 °C.

Expected Result

pressure 2.2–2.8 barg and oil 45–60 °C

Scenario 5: Liquid carryover

Field Condition

scrubber level 78 % from 35–55; suction 7.1 barg; vibration 9.0 mm/s RMS

Problem

High liquid level before the compressor can cause slugging and vibration.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses level above 65 %.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when level 35–55 % and vibration below site alarm.

Expected Result

level 35–55 % and vibration below site alarm

Scenario 6: High suction pressure

Field Condition

suction 14.2 barg from 9–11; discharge 30.4 barg; load 82 %

Problem

High suction pressure shifts the operating point and increases load.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses suction above 12 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when suction 9–11 barg with load 60–75 %.

Expected Result

suction 9–11 barg with load 60–75 %

Scenario 7: Low suction pressure

Field Condition

suction 5.4 barg; discharge 28.0 barg; flow 2.1 m³/min from 3.4

Problem

Low suction and reduced flow may indicate upstream restriction or low supply.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses suction below 7 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when suction 9–11 barg and flow 3.2–3.6 m³/min.

Expected Result

suction 9–11 barg and flow 3.2–3.6 m³/min

Scenario 8: High discharge pressure

Field Condition

discharge 36.5 barg from 28–32; load 91 %; vibration 6.2

Problem

Downstream restriction is raising head and load.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses discharge above 34 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when discharge 28–32 barg and load 60–75 %.

Expected Result

discharge 28–32 barg and load 60–75 %

Scenario 9: Anti-surge instability

Field Condition

recycle valve cycles 22–68 % every 9 s; flow 2.4 m³/min; vibration 5.8

Problem

Rapid recycle cycling indicates control tuning or bad flow/pressure input.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses cycle amplitude above 20 %.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when recycle movement below 10 % with stable flow.

Expected Result

recycle movement below 10 % with stable flow

Scenario 10: High-load vibration

Field Condition

load 88 %; vibration 8.6 mm/s RMS versus 3.0 at 55 %; bearing 79 °C

Problem

Vibration rises at high load, suggesting operating-point, piping, or rotor sensitivity.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses vibration above 7 mm/s RMS.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.

4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when load limited to approved stable range with vibration below alarm.

Expected Result

load limited to approved stable range with vibration below alarm

Scenario 11: Vibration after maintenance

Field Condition

vibration 7.4 mm/s RMS after seal replacement versus 2.6 before; speed 7,200 RPM

Problem

A post-maintenance alignment, balance, or guard issue is likely.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses increase above 3 mm/s RMS.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when vibration returns below 3 mm/s RMS.

Expected Result

vibration returns below 3 mm/s RMS

Scenario 12: Foundation looseness

Field Condition

base bolt torque loss; vertical vibration 8.1 mm/s RMS; bearing 74 °C

Problem

Loose mounting amplifies vibration and can damage the skid.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses vibration above 6 mm/s RMS.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when bolts torqued and vibration 2–4 mm/s RMS.

Expected Result

bolts torqued and vibration 2–4 mm/s RMS

Scenario 13: Piping vibration

Field Condition

discharge pipe vibration 12 mm/s RMS; compressor 4.5; pressure pulsation 1.8 bar

Problem

Piping resonance or inadequate support is transmitting force to the compressor.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses pipe vibration above 8 mm/s RMS.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when pipe vibration below 4 mm/s RMS.

Expected Result

pipe vibration below 4 mm/s RMS

Scenario 14: Seal-gas problem

Field Condition

seal gas 1.2 barg above casing versus normal 2.5–3.0; leakage detector 18 ppm from 4

Problem

Low seal-gas differential can permit process gas ingress and instability.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses differential below 2.0 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when seal-gas differential 2.5–3.0 barg and detector near baseline.

Expected Result

seal-gas differential 2.5–3.0 barg and detector near baseline

Scenario 15: Discharge pulsation

Field Condition

discharge pressure fluctuates 28.0–31.8 barg; flow 3.0–3.6 m³/min; vibration 6.1 mm/s RMS; load 72 %

Problem

Pressure pulsation with elevated vibration indicates a restriction, check-valve issue, or unstable recycle path.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses pressure swing above 2 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when pressure swing below 0.5 barg and vibration 2–4 mm/s RMS.

Expected Result

pressure swing below 0.5 barg and vibration 2–4 mm/s RMS